

$R_S^*, R_R^* \rightarrow$ Symmetric and reflexive closure.

Transitive Closure, Warshall's Algo and few Examples

1. Transitive Closure of Relation R is Smallest Transitive Relation that Contains R .

$$R_T^* = R \cup \{ (a, c) \mid (a, b) \text{ and } (b, c) \in R \}$$

$$R = \{ (1, 1), (2, 3), (3, 1) \}$$

$$\begin{aligned} R_T^* &= \{ (1, 1), (2, 3), (3, 1) \} \cup \{ (2, 1) \} \\ &= \{ (1, 1), (2, 3), (3, 1), (2, 1) \} \end{aligned}$$

* Warshall's Algo

1. Represent Relation in Matrix.
2. Let n be the No. of elements in the Set on which Relation is defined. for Warshall's Algo, there will be n steps.
3. for each Stepⁱ:
 - obtain C_i and R_i from the matrix.
 - obtain $C_i \times R_i$

- Update the Relation Matrix by Putting 1 in all Pairs which are in result of $C_i \times R_i$

4. At the end of n^{th} Step, the Matrix contains Transitive Closure.

Ex. $A = \{1, 2, 3, 4\}$
 $R = \{(2,1), (2,3), (3,1), (3,4), (4,1), (4,3)\}$
Find Transitive Closure.

$\sim n=4$

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Step-1

$$C_1 = \{2, 3, 4\}$$

$$R_1 = \emptyset$$

$$C_1 \times R_1 = \emptyset$$

Step-2

$$C_2 = \emptyset$$

$$R_2 = \{1, 3\}$$

$$C_2 \times R_2 = \emptyset$$

Step-3

$$C_3 = \{2, 4\}$$

$$R_3 = \{1, 4\}$$

$$C_3 \times R_3 = \{(2,1), (2,4), (4,1), (4,4)\}$$

~ Update my matrix

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

Step-4

$$C_4 = \{2, 3, 4\}$$

$$R_4 = \{1, 3, 4\}$$

$$C_4 \times R_4 = \{(2,1), (2,3), (2,4), (3,1), (3,3), (3,4), (4,1), (4,3), (4,4)\}$$

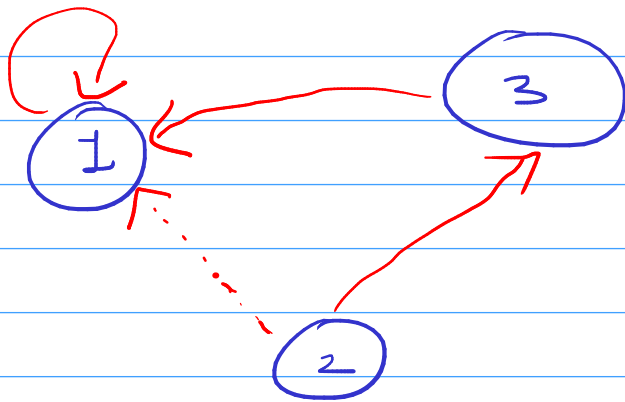
~ Update the Matrix

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

→ Transitive Closure of the Relation.

Ex.

$$R = \{(1,1), (2,3), (3,1)\}$$



~ The transitive Closure indicates Reachability in the Graph.

Ex.

$R = \{(a,b) \mid a \neq b\}$ on Set of Integers.
find Reflexive Closure.

$$R = \{(1,2), (1,3), (10,20) \dots (1,2) \checkmark, (1,1) \times\}$$

$$\begin{aligned} \sim R_R^* &= R \cup \{(a,a) \mid a \in A\} \\ &= \{(a,b) \mid a \neq b\} \cup \{(a,b) \mid a = b\} \\ &= \{(a,b) \mid a, b \in \mathbb{Z}\} \\ &= \mathbb{Z} \times \mathbb{Z} \end{aligned}$$

(5, 10) - ✓

Ex. $R = \{ (a, b) \mid a \text{ divides } b \}$
for R , find Symmetric Closure.

$$R_s^* = R \cup \{ (b, a) \mid (a, b) \in R \}$$

$$= \{ (a, b) \mid a \text{ divides } b \} \cup \{ (b, a) \mid a \text{ divides } b \}$$

$$= \{ (a, b) \mid a \text{ divides } b \} \cup \{ (a, b) \mid b \text{ divides } a \}$$

$$R_s^* = \{ (a, b) \mid a \text{ divides } b \text{ or } b \text{ divides } a \}$$

Ex. Find the Smallest Relation Containing the Relation $R = \{ (1, 2), (1, 4), (3, 3), (4, 1) \}$ that is Reflexive and Transitive.

$$\sim T = (R_R^*) \cup (R_T^*)$$

$$R_R^* = \{ (1, 2), (1, 4), (3, 3), (4, 1), (1, 1), (2, 2), (4, 4) \}$$

$$R_T^* = \{ (1, 2), (1, 4), (3, 3), (4, 1), (1, 1), (4, 4), (1, 4), ? \}$$

→ Warshall's Algo -

Ex. Find transitive Closure Using
Warshall's Algo.

$$R = \{(a, c), (b, d), (c, a), (d, b), (e, d)\}$$

$R =$

	a	b	c	d	e
a	0	0	1	0	0
b	0	0	0	1	0
c	1	0	0	0	0
d	0	1	0	0	0
e	0	0	0	1	0

- for $i = 5$

$$C_i \times R_i$$